Completing the square

1 Explain how to find the number to add to $x^2 - 5x$ to make it a perfect square trinomial.

2 Consider the expression $x^2 + 6x$.

a Does adding 36 to the expression make it a perfect square?

b What should be added to $x^2 + 6x$ to make it a perfect square?

3 Consider the equation $x^2 - 3x = 6$. What should be added to the equation in order to complete the square?

4 Complete the following expressions so they form a perfect square:

a $x^2 + 10x + \square$

b $x^2 - 2x + \square$

c $x^2 - \square x + 16$

d $x^2 - x + \square$

e $x^2 + mx + \square$

f $x^2 - \dfrac{4}{5}x + \square$

5 Complete the following perfect squares:

a $x^2 + 4x + \square = (x + \square)^2$

b $x^2 - 5x + \square = (x - \square)^2$

c $x^2 - \dfrac{7}{4}x + \square = (x - \square)^2$

6 Rewrite the following in the form $(x+b)^2 + c$ using the method of completing the square:

a $x^2 + 12x$

b $x^2 - 12x$

c $x^2 + 2x + 2$

d $x^2 - 14x + 56$

e $x^2 - 6x + 5$

f $x^2 + 3x + 6$

g $x^2 - 7x + 14$

7 Factorise the quadratics using completing the square:

a $y = x^2 + 56x + 159$

b $y = x^2 - 24x + 63$

8 Factorise the quadratics using the method of completing the square to get them into the form:

$$y = \left(x - a + \sqrt{b}\right)\left(x - a - \sqrt{b}\right)$$

a $y = x^2 - 6x + 7$

b $y = x^2 + 8x + 14$

Non-monic completing the square

9 Using the method of completing the square, rewrite $4x^2 - 36x + 79$ in the form $c\left((x+a)^2 + b\right)$.

10 Rewrite the following in the form $a(x+b)^2 + c$ using the method of completing the square:

a $3x^2 + 9x + 8$

b $4x^2 - 11x + 7$

11 Use completing the square to factorise the quadratic $y = 2x^2 + 25x + 23$.

Solving equations

12 Consider the equation $x^2 + 24x = 10$. To solve the equation by completing the square, state whether each of the following lines of working could be used:

a $x^2 + 24x + 144 = 10$

b $(x+12)^2 - 144 = 10$

c $(x+12)^2 = 10$

d $(x+12)^2 - 144 = -134$

e $x^2 + 24x + 144 = 10 + 144$

13 Solve the following quadratic equations:

a $(x+6)^2 = 4$

b $(4x-4)^2 = 4$

c $2(x-3)^2 = 8$

d $(x+2)^2 = 20$

e $(x+5)^2 - 2 = 15$

f $(x+9)^2 = \dfrac{15}{2}$

14 Solve the following quadratic equations by completing the square:

a $x^2 + 18x + 32 = 0$

b $x^2 - 6x + 8 = 0$

c $x^2 - 9x + 8 = 0$

d $2x^2 - 12x - 32 = 0$

e $4x^2 + 11x + 7 = 0$

f $x^2 - 2x - 32 = 0$

15 Solve the following quadratic equations by completing the square. Leave your answers in surd form:

a $x^2 + 24x + 5 = 0$

b $x^2 + 11x + 5 = 0$

c $6x^2 + 48x + 24 = 0$

d $x^2 - 7x + 8 = 0$

Applications

16 A circle has equation $(x+2)^2+(y-4)^2=41$. Find the x-intercepts of the circle.

17 Find the value(s) of k that will make $x^2+kx+16$ a perfect square trinomial.

18 The revenue y (in millions of dollars) of a company x years after it first started is modelled by

$$y=12.5x^2-64x+135$$

By completing the square, use this equation to predict the number of years x after which the revenue of the company will reach $\$1167$ million.